

Systems whos energy has an upper bound

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The fact that there is an upper bound is related to the appearance of negative temperature.
In this case $\exists T$ such that

$$kT \gg \frac{E}{N} \quad (1)$$

Consider canonical partition function

$$Z \approx \left(\sum_n^e -\beta \epsilon_n \right)^N \quad (2)$$

(neglecting interactions for high T) $\beta \epsilon_n \ll 1$

$$\Rightarrow Z \approx \left(\sum_n \left(1 - \beta \epsilon_n + \frac{1}{2} \beta^2 \epsilon_n^2 \right) \right)^N \quad (3)$$

$$\ln Z \approx N \ln \left[\sum_n 1 - \sum_n \beta \epsilon_n + \frac{1}{2} \sum_n \beta^2 \epsilon_n^2 \right] \quad (4)$$

To proceed we set $\sum_n 1 \approx g$ e.g. all possible orientations in a magnetic field, all possible polarization's etc.

$$\sum_n \beta \epsilon_n \approx g \beta \langle \epsilon_n \rangle \quad (5)$$

$$\sum_n \beta^2 \epsilon_n^2 \approx g \beta^2 \langle \epsilon_n^2 \rangle \quad (6)$$

Subbing into $\ln Z$.

$$\ln Z \approx N \ln \left[g \left(1 - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 \langle \epsilon_n^2 \rangle \right) \right] \quad (7)$$

$$\approx N \left[\ln g - \left(\beta \langle \epsilon_n \rangle - \frac{1}{2} \beta^2 \langle \epsilon_n^2 \rangle \right) - \frac{1}{2} \left(\beta \langle \epsilon_n \rangle - \frac{1}{2} \beta^2 \langle \epsilon_n^2 \rangle \right)^2 + \dots \right] \quad (8)$$

$$\approx N \left[\ln g - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 (\langle \epsilon_n^2 \rangle - \langle \epsilon_n \rangle^2) \right] \quad (9)$$

so we can use

$$\ln Z \approx N \left[\ln g - \beta \langle \epsilon_n \rangle + \frac{1}{2} \beta^2 (\langle \epsilon_n^2 \rangle - \langle \epsilon_n \rangle^2) \right] \quad (10)$$

into

$$\mathcal{F} = -kT \ln Z \quad (11)$$

$$\approx -NkT \ln g + N \langle \epsilon_n \rangle - \frac{N}{2kT} \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad (12)$$

$$S \approx Nk \ln g - \frac{Nk}{2} \beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad (13)$$

$$\langle E \rangle \approx N \langle \epsilon_n \rangle - N \beta \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \quad (14)$$

$$C_V \approx Nk \beta^2 \langle (\epsilon_n - \langle \epsilon_n \rangle)^2 \rangle \geq 0 \quad (15)$$

These formulae determine the thermodynamic behavior of the system, for $\beta \rightarrow 0^+$ or $\beta \rightarrow 0^-$

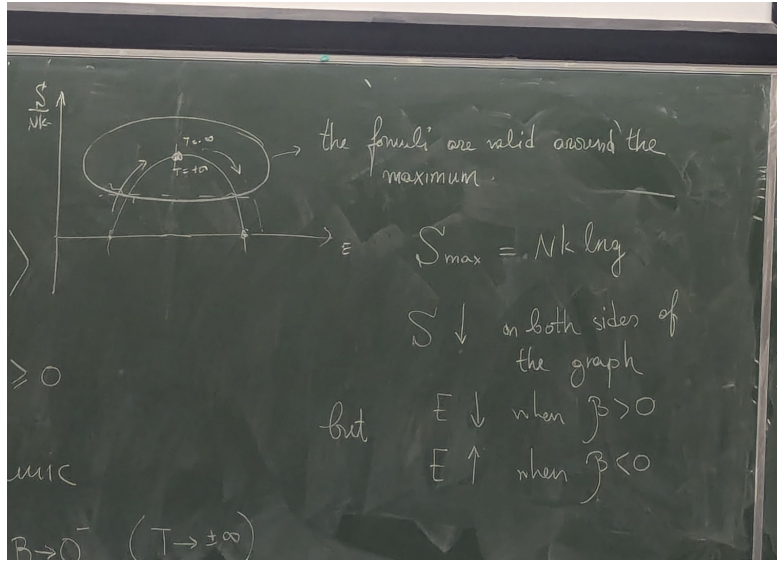


Figure 1: 4.1.png

Exercise

Ideal gas of (indistinguishable) electrons in a constant magnetic field $\vec{H}$ enclosed in a volume V .

Compute the average dipole moment and the magnetic susceptibility weak or strong magnetic fields, large or small T , Curie Law?.

$$\ln \Xi = \sum_n \ln (1 + e^{-\beta \epsilon_n} e^{\beta \mu}) \quad (16)$$

+ in log since we have fermions
We add a magnetic field so

$$e^{-\beta \epsilon_n} \rightarrow e^{-\beta \epsilon_n(p) \pm \beta \vec{\mu} H} \quad (17)$$

$$\ln \Xi = \sum_n \ln (1 + e^{-\beta \epsilon_n(p) \pm \beta \vec{\mu} H} e^{\beta \mu}) \quad (18)$$

$$= \sum_n \ln (1 + e^{-\beta \epsilon_n(p)} e^{\beta \mu_+}) + \sum_n \ln (1 + e^{-\beta \epsilon_n(p)} e^{\beta \mu_-}) \quad (19)$$

$$= \sum_n \ln (1 + z_+ e^{-\beta \epsilon_n(p)}) + \sum_n \ln (1 + z_- e^{-\beta \epsilon_n(p)}) \quad (20)$$

The net effect of introducing a magnetic field to shift a chemical potential μ Where we have defined

$$\mu_{\pm} = \mu \pm \vec{\mu} H \quad (21)$$

$$z_{\pm} = e^{\beta \mu_{\pm}} = e^{\beta (\mu \pm \vec{\mu} H)} \quad (22)$$

Where we have the $+$ $\implies$ disalignment and $-$ $\implies$ alignment

$$J = -kT \ln \Xi \approx (\text{deg}) \frac{kTV}{\lambda_T^3} \left(\mathcal{L}i_{\frac{5}{2}}(-z_+) + \mathcal{L}i_{\frac{5}{2}}(-z_-) \right) \quad (23)$$

$$\lambda_T = \left(\frac{h^2}{2\pi m kT} \right) \quad (24)$$

$$\langle M \rangle = -\frac{1}{N} \left(\frac{\partial J}{\partial H} \right)_{T, \mu, V} \quad (25)$$

$$\chi_T = \left(\frac{\partial \langle M \rangle}{\partial H} \right)_{T, V, N} \quad (26)$$

$$\langle M \rangle = \dots = \frac{\vec{\mu}}{N} (N_+ - N_-) \quad (27)$$

we can associate $\langle N_+ \rangle$ with the poly log of $-z_+$ and similarly for $-z_-$.

For small temperatures

$$\chi_T \approx -\frac{V}{N} \frac{\vec{\mu}}{\lambda_T^3} (\beta \vec{\mu}) \left[\mathcal{L}i_{\frac{1}{2}}(-z_+) + \mathcal{L}i_{\frac{1}{2}}(-z_-) \right] \approx -\frac{2\beta \vec{\mu}^2 V}{\lambda_T^3} \mathcal{L}i_{\frac{1}{2}}(-z) \quad (28)$$

Then we can show that the curie law is valid.

If we had bosons we would have 3 possible orientations 3 possible terms to add to the energy 3 polylogs. Would we have bose einstein condensation. for which orientation would be occur.